PS11 · DATA SCIENCE & ANALYTICS

StockSense

Never run out. Never overstock.



STACK LABS · K.S.R College of Engineering

Araafath S (Team Lead)

Baladith R

Gowtham R

Ajay S

STOCKSENSE · THE CORE LOOP

From five inputs to one ranked order list

01	02	03	04	05
Upload sales	Enter stock + lead time	Forecast demand	Rank quantity + urgency	Export

TOP PRIORITY

Order 48 units

Run out in 2 days

CRITICAL



Shelf-life aware



Best model per product



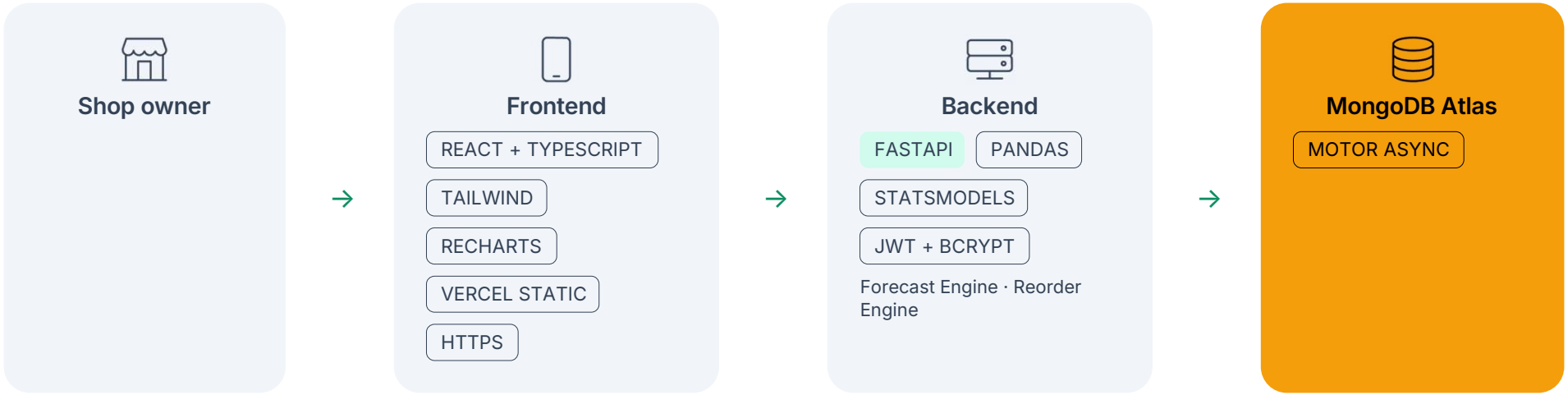
Explainable quantity



2-minute daily use

SYSTEM DESIGN

Lightweight architecture, explainable decisions



DATA PIPELINE

Sales file → Clean → Daily demand → Pick model → Forecast → Safety stock + shelf-life cap + MOQ/pack → Ranked list

01 Scaffold & Auth

02 Upload

03 Forecast

04 Reorder

05 Dashboard + deploy

Safety Stock = z × sigma × sqrt(L+R)

Priority = 0.45 Urgency + 0.35 Profit-at-risk + 0.20 Velocity

Feasible now. Scalable next.



Technical

Lightweight libraries · Serverless-ready



Operational

Sales file + a few fields · Phone-first



Economic

Vercel + MongoDB Atlas · Low running cost



Scalable

Kirana → pharmacy → bakery → multi-store

Risk	Mitigation
Short history	Confidence flag
Messy files	Column mapping + row errors
Festival spikes	Demand multiplier
Wrong forecast	Owner override + WAPE

THE OUTCOME

From memory-based ordering to confident replenishment



BEFORE

Memory · Stock-outs · Cash locked · Expiry waste





AFTER

Ranked list · Right quantities · Less waste · Minutes per order

TARGET, TO BE VALIDATED

[X]% fewer stock-outs

TARGET, TO BE VALIDATED

[X]% less expiry wastage

TARGET, TO BE VALIDATED

[X] min saved per order

Shop owner · Customers & suppliers · Community & economy

STOCKSENSE · FIVE OBJECTIVES

Order smarter.
Waste less. Sell more.

01	02	03	04	05
Forecast demand	Rank exact quantities	Respect constraints	Cut stock-outs + cash lock	Stay simple + explainable